

EXPERIMENT - 16

Title

Build a Music Store Application using React Components and provide routing among the web pages.

Aim

To develop a Music Store Web Application using:

- React Components
 - React Router DOM
 - Component-based architecture
 - Professional styling
 - Navigation between multiple pages
-

Prerequisites

- JavaScript (ES6)
 - React fundamentals
 - useState Hook
 - React Router knowledge
 - Node.js installed
-

Software Requirements

- Node.js (LTS)
 - React
 - react-router-dom
 - VS Code
 - Chrome Browser
-

Hardware Requirements

- 4GB RAM minimum
 - Intel i3+
 - 5GB free space
-

STEP 1 - Create Project

Shell

```
npx create vite@latest  
cd music-store  
npm install react-router-dom  
npm start
```

Project Structure

None

```
src/
```

```
├─ App.jsx
├─ Header.jsx
├─ Footer.jsx
├─ Home.jsx
├─ Store.jsx
├─ About.jsx
├─ NotFound.jsx
├─ index.jsx
└─ index.css
```

src/App.jsx

None

```
import React from "react";
import { BrowserRouter as Router, Routes, Route } from
"react-router-dom";

import Header from "./Header";
import Footer from "./Footer";
import Home from "./Home";
import Store from "./Store";
import About from "./About";
import NotFound from "./NotFound";

const App = () => {
  return (
    <Router>
      <Header />

      <Routes>
```

```
    <Route path="/" element={<Home />} />
    <Route path="/store" element={<Store />} />
    <Route path="/about" element={<About />} />
    <Route path="*" element={<NotFound />} />
  </Routes>

  <Footer />
</Router>
);
};

export default App;
```

src/Header.jsx

```
None
import React from "react";
import { Link } from "react-router-dom";

const Header = () => {
  return (
    <header>
      <nav>
        <ul>
          <li><Link to="/">Home</Link></li>
          <li><Link to="/store">Store</Link></li>
          <li><Link to="/about">About</Link></li>
        </ul>
      </nav>
    </header>
  );
};
```

```
    );  
  };  
  
  export default Header;
```

src/Footer.jsx

```
None  
import React from "react";  
  
const Footer = () => {  
  return (  
    <footer>  
      <p>&copy; 2024 Music Store</p>  
    </footer>  
  );  
};  
  
export default Footer;
```

src/Home.jsx

```
None  
import React from "react";  
  
const Home = () => {  
  return (  

```

```
    <main>
      <h1>Welcome to the Music Store</h1>
      <p>Your one-stop shop for musical instruments and
gear!</p>
    </main>
  );
};

export default Home;
```

src/Store.jsx

None

```
import React from "react";

const Store = () => {
  const products = [
    { id: 1, name: "Acoustic Guitar", price: "$200" },
    { id: 2, name: "Electric Guitar", price: "$400" },
    { id: 3, name: "Drum Set", price: "$600" },
  ];

  return (
    <main>
      <h1>Store</h1>
      <ul>
        {products.map((product) => (
          <li key={product.id}>
            {product.name} - {product.price}
          </li>
        ))}
      </ul>
    </main>
  );
};
```

```
        ))}
      </ul>
    </main>
  );
};

export default Store;
```

src/About.jsx

```
None
import React from "react";

const About = () => {
  return (
    <main>
      <h1>About Us</h1>
      <p>
        We are passionate about music and dedicated to
        serving musicians worldwide.
      </p>
    </main>
  );
};

export default About;
```

src/NotFound.jsx

None

```
import React from "react";

const NotFound = () => {
  return (
    <main>
      <h1>404 - Page Not Found</h1>
      <p>Sorry, the page you are looking for does not
exist.</p>
    </main>
  );
};

export default NotFound;
```

src/index.jsx

None

```
import React from "react";
import { createRoot } from "react-dom/client";
import App from "./App";
import "./index.css";

const root = createRoot(document.getElementById("root"));

root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

src/index.css

CSS

```
body {  
  font-family: Arial, sans-serif;  
  margin: 0;  
  padding: 0;  
}
```

```
header nav ul {  
  display: flex;  
  list-style: none;  
  background-color: #333;  
  padding: 0;  
  margin: 0;  
}
```

```
header nav ul li a {  
  color: white;  
  text-decoration: none;  
  padding: 10px 20px;  
  display: block;  
}
```

```
header nav ul li a:hover {  
  background-color: #575757;  
}
```

```
footer {  
  text-align: center;  
  background: #333;  
  color: white;  
  padding: 10px 0;  
  position: fixed;
```

```
    bottom: 0;
    width: 100%;
  }

  main {
    padding: 20px;
    margin-bottom: 60px;
  }
```

Features Implemented

Feature	Status
React Components	✓
React Router	✓
Navigation	✓
Dynamic Store Page	✓
Professional Styling	✓
Reusable Components	✓

Expected Result

The Music Store application successfully:

- Navigates between pages using routing
 - Displays music items
 - Adds items to cart
 - Uses modern UI styling
-

Result

Thus, the Music Store Application was successfully developed using React components and routing with professional styling.